Effects of waterlogging on the yield and growth of summer maize under field conditions

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Ren, B., Zhang, J., Li, X., Fan, X., Dong, S., Liu, P. and Zhao, B. 2014. **Effects of waterlogging on the yield and growth of summer maize under field conditions**. Can. J. Plant Sci. **94**: 23–31. A field experiment was performed to study the effects of waterlogging for different durations (3 and 6 d) on the yield and growth of summer maize at the three-leaf stage (V3), six-leaf stage (V6), and the 10th day after the tasseling stage (10VT). The results after 2 yr indicated that maize development and grain yield responses to waterlogging depended on both stress severity (intensity and duration) and different growth stage. Yield decreased significantly with an increased waterlogging duration during V3 and V6. The yields of maize hybrid Denghai 605 (DH605) in treatments V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 were 23, 32, 20, 24, 8, and 18% lower than those of the control (CK), respectively; Yields of Zhengdan 958 (ZD958) were lower by 21, 35, 15, 33, 7, and 12%, respectively. Waterlogging also affected the growth and development of summer maize. Ear characteristics (grains per ear and 1000-grain weight) and plant morphology (plant height, ear height, and leaf area index) decreased, whereas the bald tip length increased significantly. The maximum grain-filling rate decreased under waterlogging; furthermore, the dry matter accumulation decreased and dry matter distribution proportions of the stem and leaf increased. However, the distribution proportion of grain decreased. Maize was most susceptible to waterlogging damage at V3, followed by V6 and 10VT, with damage increasing with increasing waterlogging duration.

Key words: Summer maize, waterlogging, grain yield, grain filling, dry matter accumulation and distribution, plant characteristics

Ren, B., Zhang, J., Li, X., Fan, X., Dong, S., Liu, P. et Zhao, B. 2014. **Incidence de l'engorgement du sol sur le rendement et la croissance du maïs au champ**. Can. J. Plant Sci. **94**: 23–31. On a procédé à une expérience sur le terrain pour étudier les effets d'un engorgement de durée variable (3 et 6 jours) sur le rendement et la croissance du maïs au stade de la troisième feuille (V3), à celui de la sixième feuille (V6) et 10 jours après la floraison mâle (10VT). Les résultats obtenus au bout de deux ans indiquent que le développement de la plante et le rendement grainier réagissent à l'ennoyage en fonction de la gravité du stress (intensité et durée) et du stade de croissance. Le rendement diminue significativement quand l'engorgement dure plus longtemps aux stades V3 et V6. Le rendement du maïs hybride Denghai 605 (DH605) lors des traitements V3-3, V3-6, V6-3, V6-6, 10VT-3 et 10VT-6 était respectivement de 23 %, 32 %, 20 %, 24 %, 8 % et 18 % inférieur à celui du cultivar témoin (CK); le rendement de la variété Zhengdan 958 (ZD958) était pour sa part plus faible de 21 %, 35 %, 15 %, 33 %, 7 % et 12 %, respectivement. L'ennoyage touche aussi la croissance et le développement du maïs. Ainsi, on note une détérioration des caractéristiques de l'épi (nombre de grains par épi et poids de 1 000 grains) et de la morphologie de la plante (hauteur du plant, hauteur de l'épi, indice de la surface foliaire), alors que la partie nue à l'extrémité s'allonge sensiblement. Le taux maximal de remplissage du grain diminue quand il y a engorgement du sol; parallèlement, la matière sèche s'accumule moins et sa proportion augmente dans la tige et les feuilles. Toutefois, on remarque une baisse de la proportion de matière sèche dans le grain. Le maïs s'avère particulièrement sensible à l'ennoyage au stade V3, puis aux stades V6 et 10VT, les dommages s'accroissant avec la durée de l'engorgement du sol.

Mots clés: Maïs, engorgement, rendement grainier, remplissage du grain, accumulation de matière sèche, répartition de la matière sèche, paramètres de la plante

Waterlogging is considered to be one of the major problems in crop production, affecting an estimated 12% of the global cultivated area (Li 1997). Waterlogging decreases crop yield (Bhan 1977; Belford 1981; Bange et al. 2004; Dickin and Wright 2008). Previous studies have reported various negative effects of waterlogging on crops. For example, increasing durations of waterlogging decreased maize yield (Yang and Chen 1998; Przywara and Stepniewski 1999; Li et al. 2001; Milroy et al. 2009; Araki et al. 2012). Waterlogging also significantly affects plant morphology, decreasing cell permeability (Krause and Weiss 1984; Patwardhan et al.

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Abbreviations: **CK**, control; **DH605**, Denghai 605; **LAI**, leaf area index; **R3**, milk ripe stage; **R6**, physiological maturity stage; **V3**, three-leaf stage; **V6**, six-leaf stage; **VT**, tasseling stage; **10VT**, 10th day after the tasseling stage; **ZD958**, Zhengdan 958

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1988; Musgrave and Ding 1998; Tan et al. 2007), reducing root activity and root respiration, and accelerating root senescence (Mckevlin et al. 1998; Grzesiak et al. 1999; Li et al. 2001; Malik et al. 2001; Li et al. 2011). Maize requires large amounts of water, but is not resistant to waterlogging. Chen et al. (1988) found that when soil moisture reached more than 80% of field capacity, maize growth and development were greatly reduced. In the Huanghuaihai Plain of China, the most rainfall occurs during the growing season of summer maize, and growth and yield of maize are significantly affected by farmland water storage. The effect of waterlogging on maize varies among cultivars, growth stages, environmental conditions, and waterlogging duration. Previous studies have reported decreases in grain yield of 20–35% after waterlogging for only 1–2 d at V3 (Ritter and Beer 1969; Howell and Hiler 1974; Bhan 1977). Bhan (1977) reported that no differences existed between waterlogging for 5 d at V3 and at the tasseling stage (VT), whereas Ritter and Beer (1969) reported that waterlogging at VT did not affect yield. Chen et al. (1989) reported that the negative impacts of waterlogging varied with growth stage; early stages were the most susceptible to waterlogging, and waterlogging for 3 d at V3, V6, and the differentiation period of small female ears resulted in 13.2, 16.2, and 7.9% decreases in yield compared with the control, respectively. However, waterlogging for 3 d at VT or during the initial milky stage did not affect grain yield (Chen et al. 1989). Li et al. (2011) found that waterlogging for 1 d had little effect on maize production, but waterlogging for more than 3 d resulted in a more than 40% reduction in grain yield; At V6, waterlogging for 5–7 d (as well as 7 d at VT) prevented harvest of summer maize.

Previous studies have shown that waterlogging can decrease maize yield. However, results have varied because of the different materials and methods used. Furthermore, most previous studies have been performed with pot experiments. Plants grown in pots have restricted spatial and root growth, which limits the natural responses of plant growth. In addition, few studies have assessed the interaction of waterlogging stage and duration on the ear trait, grain filling, growing periods and dry matter distribution. The objective of this study was to determine the effect of waterlogging for 3 or 6 d on yield and growth of field-grown summer maize at the three-leaf stage (V3), six-leaf stage (V6), and the 10th day after the tasseling stage (10VT).

MATERIALS AND METHODS

Plant Materials and Experimental Design

The field experiment was conducted during 2011 and 2012 at the State Key Laboratory of Crop Biology and the experimental farm of Shandong Agricultural University, China (lat. 36°10′N, long. 117°04′E, 151 m a.s.l.). The region has a temperate continental monsoon climate. The experimental soil type is brown loam. Concentrations of organic matter, total N, rapidly available phosphorus (P), and rapidly available potassium (K) in the upper 20 cm of soil were 10.71 g kg^{-1}, 0.89 g kg^{-1}, 50.65 mg kg^{-1}, and 86.15 mg kg^{-1}, respectively. Each plot was 4 m × 4 m and separated by 4 m × 2.3 m polyvinyl chloride (PVC) boards as water barriers. Every PVC board was buried 2.0 m below the surface and the remaining 0.3 m was above ground. The summer maize hybrids DengHai605 (DH605) and ZhengDan958 (ZD958), which are planted most popularly in China, were used for experimental materials. Maize was sown on Jun. 16 in both years with a plant density of 67 500 plants ha^{-1}. Details of the experimental treatments are shown in Table 1. Disease, weeds, and pests were well controlled in each treatment. Herbicide of 90% atrazine acetochlor was diluted 2000–3000 times and sprayed on the whole field surface at the rate of 600 L ha^{-1} before germinating to control weeds; pesticide of 50% phoxim emulsifiable concentrate were diluted 1000 times with water and sprayed at the rate of 750 L ha^{-1} at the nine-leaf stage (V9) to control corn borers. During the waterlogging treatments, the water in the waterlogged main pools was maintained at 2-3 cm above the soil surface. Each treatment was replicated three times, in a completely randomized design.

Table 1. Waterlogging treatments in the field from 2011 to 2012

Year	Cultivar	Treatment	Year	Cultivar	Treatment
2011	DH605 ZD958	– Waterlogging for 3 d at jointing stage (V6-3) Waterlogging for 6 d at jointing stage (V6-6) Waterlogging for 3 d at the 10th day after tasseling stage (10VT-3) Waterlogging for 3 d at the 10th day after tasseling stage (10VT-6) CK	2012	DH605 ZD958	Waterlogging for 3 d at three-leaf stage (V3-3) Waterlogging for 6 d at three-leaf stage (V3-6) Waterlogging for 3 d at jointing stage (V6-3) Waterlogging for 6 d at jointing stage (V6-6) Waterlogging for 3 d at the 10th day after tasseling stage (10VT-3) Waterlogging for 6 d at the 10th day after tasseling stage (10VT-6) CK

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Dry Matter Accumulation

Five representative plant samples were obtained from each plot during the six-leaf stage (V6), twelve-leaf stage (V12), tasseling stage (VT), milk ripe stage (R3), and physiological maturity stage (R6). Samples were preserved after being separated into stems and leaves at V6, V12, and VT, and into stems, leaves, and ears at R3 and R6. Samples were dried at 80°C in a force-draft oven (DHG-9420A, Shanghai Bilon Instruments Co. Ltd. of China) to constant weight and weighed separately.

Plant Morphology

Fifteen representative plant samples were marked from each plot at V6, V12, VT, R3, and R6 to measure leaf length and width, plant height, ear height. Leaf area, leaf area index (LAI), and ear position coefficients were calculated as:

$$\text{Leaf area} = \text{leaf length} \times \text{leaf width} \times 0.75$$

$$\text{LAI} = (\text{leaf area per plant} \times \text{stock number per plot})/\text{plot area}$$

$$\text{Ear position coefficients} = \text{ear height}/\text{plant height} \times 100\%$$

Grain-filling Characteristics

From pollination to R6, 100 grains from the middle and lower portions of five ears open- pollinated on the same day were obtained to determine the dry weight of grain per 5 d. Logistic equations ($y = a/(1+be^{-cx})$) simulating the grain-filling process were fitted with CurveExpert1.3 to measure days of maximum grain filling (Dmax), weight of maximum grain filling rate (Wmax), maximum grain filling (Gmax), and active grain filling period (P).

$$\text{Dmax} = \ln(b/c);\ \text{Wmax} = 0.5 \times a;$$

$$\text{Gmax} = (c \times \text{Wmax}) \times [1 - (\text{Wmax}/a)];\ \text{P} = 6/c.$$

where a represents the ultimate growth, b represents the initial parameters, and c represents the growth rate parameters.

Grain Yield

At R6, 30 ears harvested from three rows at the center of each plot were used to determine yield (moisture content was approximately 14%) and ear traits including length, width, weight, row number, kernels per row, bald tip length, cob weight, and cob width.

Statistical Analysis

Comparisons among different treatments were based on Duncan's test at the 0.05 probability level ($P \leq 0.05$). Analysis of variance (ANOVA) was performed for grain yield, dry matter weight, ear trait, and plant characteristics using SPSS17.0 (SPSS Institute Inc.). Similar data, with only slight variation were found in 2011 and 2012. In this paper, priority is given to the analysis of 2012 data, which had more treatments.

RESULTS

Grain Yield

Yield significantly decreased with increased waterlogging duration at different growth periods. In 2012, grain yields of DH605 in treatments V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 were 23, 32, 20, 24, 8 and 18% lower than in the CK, respectively. ZD958 yields in treatments V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 were lower than CK yields by 21, 35, 15, 33, 7, and 12%, respectively. These results indicate that the two cultivars responded similarly to waterlogging, and were most susceptible to damage at V3, followed by V6 and 10VT. Furthermore, yield reductions increased with increasing waterlogging duration; however, yield reductions in DH605 were smaller than those in ZD958 after waterlogging for 6 d at V3 and V6, whereas for all other treatments, yield reductions in DH605 were larger than those in ZD958. In addition, grains per ear of DH605 waterlogged at V3 and V6 were reduced compared with CK, with the most significant reduction in V3-6 with a decrease of 17%. No significant differences existed between waterlogging treatments. Grains per ear waterlogged at 10VT also decreased compared with CK, but no significant differences occurred between 10VT-3 and CK. The effects on grains per ear were similar between ZD958 and DH605. The maximum reduction of grains per ear in V3-6 of ZD958 was 20%. No differences occurred between 10VT and CK. However, waterlogging at V3 and V6 significantly decreased the grains per ear of ZD958, and the effect of waterlogging at V3 was greater than that at V6. The 1000-grain weight decreased after waterlogging at different growth stages, and differed between the cultivars. The 1000-grain weight of DH605 in treatments V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 decreased by 10, 12, 6, 9, NS, and 8%, respectively. No significant difference existed between 10VT-3 and CK. The 1000-grain weights of ZD958 in treatments V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 were 10, 12, 8, 13, NS, and 5% lower than in CK, but no differences occurred between different waterlogging durations at V3 and 10VT. Under waterlogging stress, the percent of barrenness increased, resulting in fewer harvested ears. The greatest percent of barrenness of the two cultivars occurred in V3-6, with 8.57% in DH605 and 9.35% in ZD958. Similar results were found in both 2011 and 2012 (Table 2).

Ear Traits

Table 3 shows that waterlogging affected the ear length of summer maize. The ear lengths of DH605 in V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 were 6%, 8%, 6%, 7%, NS, and NS lower than CK values, respectively. Differences existed between CK and waterlogging at V3 and V6, and the effects of V3-6 were greatest, with

Table 2. Effect of waterlogging on grain yield and yield components of summer maize

Year	Cultivar	Treatment	Harvest ear number (ears ha^{-1})	Grains per ear	1000-grain weight (g)	Barrenness (%)	Yield (kg ha^{-1})	Variation amplitude (%)
2012	DH605	V3-3	63 125*c*	474*bcd*	363*c*	6.48	10 765*e*	−23.20
		V3-6	61 714*d*	438*d*	354*c*	8.57	9 512*g*	−32.14
		V6-3	63 818*bc*	466*bcd*	379*b*	5.45	11 262*d*	−19.66
		V6-6	61 822*d*	459*cd*	366*c*	8.41	10 591*f*	−24.45
		10VT-3	64 375*b*	500*ab*	402*a*	4.63	12 961*b*	−7.54
		10VT-6	63 851*bc*	480*bc*	373*bc*	5.41	11 437*c*	−18.41
		CK	65 642*a*	530*a*	403*a*	2.75	14 018*a*	
	ZD958	V3-3	64 255*bc*	522*b*	329*c*	4.81	10 943*e*	−21.33
		V3-6	61 192*e*	453*c*	322*c*	9.35	9 012*g*	−35.22
		V6-3	63 000*d*	558*ab*	337*bc*	6.67	11 770*d*	−15.39
		V6-6	62 100*e*	481*c*	317*c*	8.00	9 292*f*	−33.20
		10VT-3	64 375*b*	569*a*	358*ab*	4.63	12 951*b*	−6.89
		10VT-6	63 367*cd*	543*ab*	349*b*	6.12	12 173*c*	−12.49
		CK	66 250*a*	573*a*	366*a*	1.85	13 910*a*	
2011	DH605	V6-3	65 045*abc*	559*ab*	314*c*	3.64	11 438*d*	−11.57
		V6-6	63 978*c*	521*b*	304*d*	5.22	10 137*e*	−21.63
		10VT-3	65 172*ab*	581*a*	322*b*	3.45	12 188*b*	−5.77
		10VT-6	64 375*bc*	553*ab*	312*cd*	4.63	11 106*c*	−14.14
		CK	65 607*a*	588*a*	335*a*	2.80	12 935*a*	
	ZD958	V6-3	63 851*c*	528*ab*	320*b*	5.41	10 795*c*	−10.19
		V6-6	64 432*bc*	507*b*	302*c*	4.55	9 869*d*	−17.89
		10VT-3	65 692*a*	538*ab*	326*ab*	2.68	11 528*b*	−4.09
		10VT-6	64 800*ab*	519*ab*	323*b*	4.00	10 858*c*	−9.67
		CK	65 676*a*	550*a*	333*a*	2.70	12 020*a*	

a–g Means followed by the same letter within a column do not differ significantly at $P < 0.05$ according to Duncan's Multiple Range Test.

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no obvious differences between waterlogging treatments. ZD958 decreased by 8, 13, NS, 11, NS, and 6%, respectively. There were no significant differences between the two durations at 10VT. Waterlogging had similar effects on the ear weights of the two cultivars. The decreasing order of ear weights was CK, 10VT-3, 10VT-6, V6-3, V3-3, V6-6, and V3-6, but no differences occurred between V6-3 and 10VT-6 (or between V3-3 and V6-6). After waterlogging, the bald tip length of summer maize increased significantly. The bald tip lengths of DH605 in V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 were 22, 67, 45, 37, and 52% greater than those of CK, respectively. The bald tip lengths of ZD958 were 54, 129, 47, 76, 21, and 26% greater, respectively, than those of CK. No differences existed among the treatments of waterlogging for 3 d during the growth stages and between waterlogging at 10VT and CK. Waterlogging did not affect ear width among the waterlogging treatments or of waterlogging for 3 d at various growth stages on cob weight. Waterlogging for

Table 3. Effect of waterlogging on ear trait of summer maize

Cultivar	Treatment	Ear length (cm)	Ear width (cm)	No. of rows per ear	No. of kernels per row	Bald tip length (cm)	Ear weight (g)	Cob weight (g)	Cob width (cm)
DH605	V3-3	19.30*bc*	5.10*bc*	16*b*	30*a*	3.18*cd*	144.50*d*	24.67*abc*	2.81*b*
	V3-6	18.90*c*	5.03*c*	16*b*	27*b*	4.33*a*	120.10*e*	22.55*c*	2.82*b*
	V6-3	19.33*bc*	5.13*ab*	18*a*	27*b*	3.76*abc*	160.49*c*	25.01*abc*	2.82*b*
	V6-6	19.06*c*	5.10*bc*	16*b*	28*b*	3.81*abc*	133.30*de*	23.73*bc*	2.99*a*
	10VT-3	20.38*a*	5.20*a*	16*b*	31*a*	3.57*bc*	195.05*b*	29.19*ab*	2.81*b*
	10VT-6	19.94*ab*	5.13*ab*	17*ab*	29*ab*	3.95*ab*	176.88*c*	28.98*ab*	2.99*a*
	CK	20.56*a*	5.20*a*	16*b*	33*a*	2.60*d*	226.00*a*	29.82*a*	2.79*b*
ZD958	V3-3	16.87*cd*	5.07*bc*	15*a*	34*b*	1.08*bc*	150.67*d*	22.18*ab*	2.88*bc*
	V3-6	15.88*e*	5.03*c*	15*a*	31*c*	1.60*a*	122.56*e*	20.10*b*	2.82*cd*
	V6-3	17.54*ab*	5.13*b*	15*a*	36*ab*	1.03*bcd*	168.88*c*	22.89*ab*	2.75*d*
	V6-6	16.28*de*	5.00*c*	15*a*	32*c*	1.23*b*	143.26*d*	21.33*b*	2.97*ab*
	10VT-3	17.61*ab*	5.30*a*	15*a*	37*a*	0.85*cd*	190.46*b*	23.32*ab*	2.98*ab*
	10VT-6	17.13*bc*	5.23*a*	15*a*	36*ab*	0.88*cd*	175.83*c*	20.05*b*	3.01*a*
	CK	18.31*a*	5.23*a*	15*a*	37*a*	0.70*d*	225.00*a*	26.33*a*	2.88*bc*

a–e Means followed by the same letter within a column do not differ significantly at $P < 0.05$ according to Duncan's Multiple Range Test.

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6 d showed obvious effects, for DH605, the cob weights of V3-6, V6-6 and 10VT-6 were reduced by 24%, 20% and NS. ZD958 decreased by 24, 19 and 24%. However, no differences occurred among waterlogging treatments. Cob width was not affected by waterlogging at V3, but cob widths of V6-6 and 10VT-6 were greater than those of CK. Waterlogging did not affect the number of rows per ear for either cultivar. Increased waterlogging duration decreased the number of kernels per row of DH605, by 9, 17, 19, and 15% in V3-3, V3-6, V6-3, and V6-6, respectively. For ZD958, the kernels per row of V3-3, V3-6, V6-3, and V6-6 were reduced by 7, 16, NS, and 13%, respectively. Waterlogging did not affect the number of kernels per row at 10VT in either cultivar.

Grain-filling Characteristics

Waterlogging affected grain-filling characteristics (Table 4). Among the treatments on DH605, the grain-filling parameters of CK were greatest, whereas those in V3-6 were lowest. Days of maximum grain filling (Dmax), maximum grain filling (Gmax), weight of maximum grain filling rate (Wmax), and active grain filling period (P) of V3-6 decreased by 25, 18, 48, and 36%, respectively. A similar tendency was observed in ZD958. After waterlogging, Dmax, Gmax, Wmax, and P of grains decreased as waterlogging duration increased. The greatest losses from waterlogging occurred at V3, followed by V6 and 10VT.

Dry Matter Accumulation and Distribution

The effects of waterlogging on dry matter accumulation were similar between the two cultivars (Fig. 1). The negative effects of waterlogging varied with the growth stage and duration of waterlogging. Dry matter accumulation and accumulation rates were significantly reduced after plants were waterlogged at various growth stages. The most significant effect of waterlogging was observed at V3, followed by V6 and 10VT. Dry matter accumulation reduced significantly with increasing waterlogging durations. Waterlogging also affected the ratio of dry matter accumulation from VT to R6 compared with the total accumulation. Waterlogging at V3 and V6 significantly affected the dry matter accumulation before VT at V3-3, V3-6, V6-3, and V6-6 in DH605, resulting in 25, 33, 11 and 19% decreases compared with CK, respectively. Dry matter accumulation of ZD958 decreased by 22, 39, 16 and 24%, respectively, compared with CK. Meanwhile, waterlogging at various stages reduced dry matter accumulation after VT. Dry matter accumulation in V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 in DH605 decreased by 34, 22, 36, 18, and 27%, respectively. ZD958 decreased by 32, 37, 22, 36, 18, and 27%, respectively. Waterlogging at 10VT significantly reduced the ratio of dry matter accumulation from VT to R6 compared with the total accumulation, which also decreased with increasing durations of waterlogging. For both cultivars, the ratio of dry matter accumulation in 10VT-6 was the smallest, decreasing by 9% in DH605 and 7% in ZD958.

Waterlogging stress affected dry matter partitioning indices, but did not change the order of allocation proportions (Table 5) that the allocation proportions of the stem and ear were larger but the allocation proportion of leaf was least. The proportion of the leaf and stem dry matter to the whole dry matter increased after waterlogging. These results indicate that waterlogging had positive effects on leaves and stems. The proportions of leaves and stems varied with the growth stage and waterlogging duration. Meanwhile, the DH605 ratio of the ear dry matter to whole dry matter decreased after waterlogging, with V3-3, V3-6, V6-3, V6-6, 10VT-3, and 10VT-6 being 7.47, 9.18, 6.83, 8.55, 4.66, and 4.54% lower than CK, respectively. Thus, waterlogging

Table 4. Effect of waterlogging on grain filling parameters of summer maize

Cultivar	Treatment	Growth curve parametric equation	Correlation coefficients	A[z]	B[z]	C[z]	Dmax[z] (d)	Wmax[z] (g 100 kernels^{-1})	Gmax[z] (g 100 kernels^{-1} d^{-1})	P[z] (d)
DH605	V3-3	$y=32.95/(1+44.63e^{-0.15x})$	0.992	32.95	44.63	0.15	26.10*b*	16.48*cd*	1.20*bc*	41.22*b*
	V3-6	$y=21.44/(1+79.31e^{-0.21x})$	0.993	21.44	79.31	0.21	20.44*d*	10.72*e*	1.15*c*	28.04*d*
	V6-3	$y=29.24/(1+67.45e^{-0.16x})$	0.997	29.24	67.45	0.16	26.13*b*	14.62*d*	1.18*c*	37.23*c*
	V6-6	$y=31.45/(1+51.05e^{-0.16x})$	0.998	31.45	51.05	0.16	24.28*c*	15.73*d*	1.27*bc*	37.04*c*
	10VT-3	$y=35.13/(1+53.24e^{-0.15x})$	0.994	35.13	53.24	0.15	27.06*a*	17.56*bc*	1.29*b*	40.84*b*
	10VT-6	$y=36.44/(1+57.88e^{-0.15x})$	0.998	36.44	57.88	0.15	27.51*a*	18.22*b*	1.34*ab*	40.68*b*
	CK	$y=41.40/(1+41.06e^{-0.14x})$	0.997	41.40	41.06	0.14	27.32*a*	20.70*a*	1.41*a*	44.12*a*
ZD958	V3-3	$y=25.81/(1+54.68e^{-0.17x})$	0.991	25.81	54.68	0.17	23.77*c*	12.90*c*	1.09*b*	35.64*e*
	V3-6	$y=25.47/(1+64.87e^{-0.18x})$	0.998	25.47	64.87	0.18	22.87*c*	12.73*c*	1.16*ab*	32.88*f*
	V6-3	$y=35.33/(1+50.02e^{-0.14x})$	0.997	35.33	50.02	0.14	28.39*a*	17.66*a*	1.22*a*	43.53*bc*
	V6-6	$y=31.58/(1+53.95e^{-0.15x})$	0.994	31.58	53.95	0.15	26.08*b*	15.79*b*	1.21*a*	39.23*d*
	10VT-3	$y=30.64/(1+37.10e^{-0.14x})$	0.995	30.64	37.10	0.14	25.54*b*	15.32*b*	1.08*b*	42.41*cd*
	10VT-6	$y=31.46/(1+30.75e^{-0.13x})$	0.997	31.46	30.75	0.13	25.67*b*	15.73*b*	1.05*b*	44.96*bc*
	CK	$y=37.83/(1+32.42e^{-0.12x})$	0.995	37.83	32.42	0.12	27.95*a*	18.91*a*	1.18*a*	48.21*a*

[z]A, B, C, equation parameters; Dmax: days of maximum grain filling; Wmax: weight of maximum grain filling rate; Gmax: maximum grain filling; P: active grain filling period.

a–f Means followed by the same letter within a column do not differ significantly at $P<0.05$ according to Duncan's Multiple Range Test.

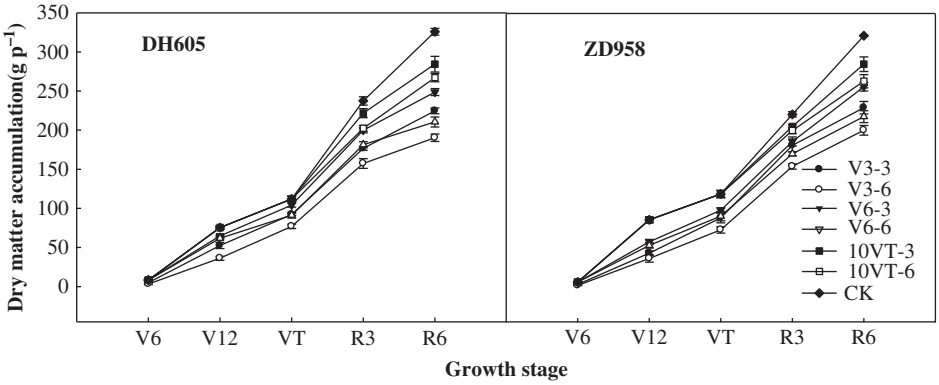


Fig. 1. Effects of waterlogging on dry matter accumulation of summer maize.

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in early stages had a greater effect than waterlogging at later stages, and the greatest effect was observed in treatment V3-6 with a decrease of 9.18%. A similar tendency occurred in ZD958, but the impact was relatively less.

Harvest index decreased after waterlogging at different growth stages and decreased with increasing waterlogging durations (excluding 10VT-3 and 10VT-6) (Table 5). Waterlogging appeared to delay maize growth and development, with the delay increasing with increasing waterlogging durations. The growth process in V3-6 was longer than that in CK by 3 d, while V3-3 and V6-6 were delayed by 2 d and V6-3, 10VT-3, and 10VT-6 were delayed by 1 d (Table 6).

Plant Morphology

Plant height was reduced with waterlogging, with DH605 showing reductions of 6, 9, 4, and 3% in treatments V3-3, V3-6, V6-6, and 10VT-6, respectively; no differences occurred between V6-3, 10VT-3, and CK. In ZD958, waterlogging at V3-3, V3-6, and V6-6 decreased plant height by 5, 14, and 8%, respectively, compared with CK, while no differences existed among V6-3, 10VT-3, and 10VT-6 and CK. In DH605, waterlogging in V3-3, V3-6, V6-3, and V6-6 decreased ear height by 5, 18, 4, and 12%, respectively, compared with CK. ZD958 experienced 7, 18, 4 and 18% decreases in ear height, respectively. No differences exised between V3-3 and V6-3, V3-6, and V6-6, indicating that ear height was not significantly affected by waterlogging treatments at V3 and V6, regardless of duration. Moreover, waterlogging at 10VT did not affect ear height (Table 7).

The LAI of both cultivars changed significantly after waterlogging (Fig. 2). For all waterlogging treatments, LAI were lower than those in CK at the corresponding growth stages and decreased with increased waterlogging durations. The overall trend of LAI was consistent

Table 5. Dry matter and its distribution of summer maize at maturity

			Stem		Leaf		Ear		
Cultivar	Treatment	Total dry matter (g p^{-1})	Dry matter (g p^{-1})	Proportion of the total (%)	Dry matter (g)	Proportion of the total (%)	Dry matter (g)	Proportion of the total (%)	Harvest index
DH605	V3-3	224.72*d*	51.63*c*	22.98	28.59*c*	12.72	24.67*bc*	10.98	0.53
	V3-6	190.27*e*	45.65*d*	23.99	24.52*d*	12.89	22.55*c*	11.85	0.51
	V6-3	248.69*cd*	57.10*b*	22.96	31.10*bc*	12.51	25.01*b*	10.06	0.54
	V6-6	210.50*de*	50.20*c*	23.85	27.00*cd*	12.83	23.73*bc*	11.27	0.52
	10VT-3	284.35*b*	62.10*a*	21.84	34.20*ab*	12.03	30.19*a*	10.62	0.56
	10VT-6	267.15*bc*	59.40*ab*	22.23	33.88*b*	12.68	26.98*ab*	10.10	0.56
	CK	325.85*a*	63.65*a*	19.53	36.20*a*	11.11	29.82*a*	9.15	0.60
ZD958	V3-3	229.67*d*	50.53*c*	22.00	28.48*cd*	12.40	20.18*bc*	8.78	0.57
	V3-6	202.61*e*	45.60*d*	22.51	25.45*d*	12.56	19.10*c*	9.43	0.56
	V6-3	256.93*c*	56.23*b*	21.88	30.83*bc*	12.00	22.89*ab*	8.91	0.58
	V6-6	219.19*de*	48.93*cd*	22.32	27.00*c*	12.32	19.33*c*	8.82	0.57
	10VT-3	285.21*b*	60.85*a*	21.34	33.90*ab*	11.89	23.32*ab*	8.18	0.59
	10VT-6	263.51*bc*	57.10*ab*	21.67	31.58*ab*	11.98	22.05*bc*	8.37	0.59
	CK	321.56*a*	61.56*a*	19.14	35.00*a*	10.88	26.33*a*	8.19	0.62

a–e Means followed by the same letter within a column do not differ significantly at $P < 0.05$ according to Duncan's Multiple Range Test.

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Table 6. Growing periods of summer maize in different treatments

Treatment	Seeding	V6	VT	R3	R6
V3-3	Jun. 16	Jul. 14	Aug. 09	Sep. 05	Oct. 05
V3-6	Jun. 16	Jul. 15	Aug. 10	Sep. 06	Oct. 05
V6-3	Jun. 16	Jul. 12	Aug. 08	Sep. 04	Oct. 05
V6-6	Jun. 16	Jul. 12	Aug. 09	Sep. 05	Oct. 05
10VT-3	Jun. 16	Jul. 12	Aug. 07	Sep. 04	Oct. 05
10VT-6	Jun. 16	Jul. 12	Aug. 07	Sep. 04	Oct. 05
CK	Jun. 16	Jul. 12	Aug. 07	Sep. 03	Oct. 05

among treatments; the LAI increased initially and then decreased during the growth period. However, waterlogging resulted in a decreased accumulation of LAI early, compared with CK, and also a more rapid loss of LAI than CK later in the season.

DISCUSSION

Previous studies demonstrated that grain yield decreased by 20–35% after waterlogging for only 1–2 d at V3 (Ritter et al. 1969; Howell et al. 1974; Bhan 1977). Bhan (1977) reported that no differences existed between waterlogging for 5 d at VT and at V3. Li et al. (2011) found that waterlogging for 1 d had little effect on maize production, but waterlogging for more than 3 d decreased yield by over 40%; At V6, waterlogging for 5–7 d, as well as 7 d at VT, resulted in total loss of summer maize. However, previous studies were not consistent and were typically conducted as pot experiments. Plants grown in pots have restricted spatial and root growth, and cannot fully represent the effects of waterlogging on summer maize under field conditions. The experiments presented here were conducted in field-flooded conditions, which could objectively measure the effects of waterlogging on maize growth and development in a more natural environment. The results indicate that maize was most susceptible to damage when waterlogging occurred at V3, followed by V6 and 10VT. These results differ from those of Chen et al. (1989), who reported that waterlogging at V6 had the greatest impact on summer maize yield. Maize grain yield decreased with increased waterlogging durations. Grain yield reductions with waterlogging for 6 d at V3 were 32% in DH605 and 35% in ZD958; these findings differed from previous studies in which waterlogging for more than 3 d decreased summer maize production by 40 to 100% (Ritter and Beer 1969; Li et al. 2011).

Table 7. Effect of waterlogging on plant height and ear height of summer maize

Cultivar	Treatment	Plant height (cm)	Ear height (cm)	Ear position coefficients
DH605	V3-3	240.8*d*	94.0*bc*	0.39*b*
	V3-6	232.7e	81.7*d*	0.30*d*
	V6-3	251.4*abc*	95.2*bc*	0.38*bcd*
	V6-6	246.1*cd*	88.0*cd*	0.36*cd*
	10VT-3	253.2*ab*	98.3*ab*	0.36*cd*
	10VT-6	248.7*bc*	104.5*a*	0.42*a*
	CK	256.6*a*	99.5*ab*	0.39*bc*
ZD958	V3-3	235.8*b*	99.6*c*	0.42*ab*
	V3-6	213.5*d*	87.7*d*	0.41*bc*
	V6-3	244.2*a*	103.4*bc*	0.42*ab*
	V6-6	228.8*c*	88.8*d*	0.39*c*
	10VT-3	247.2*a*	107.6*ab*	0.44*a*
	10VT-6	246.6*a*	109.4*a*	0.44*a*
	CK	248.6*a*	107.6*ab*	0.43*ab*

a–d Means followed by the same letter within a column do not differ significantly at $P < 0.05$ according to Duncan's Multiple Range Test.

Our results show that waterlogging at different growth stages affected ear formation, and reduced ear volume, which decreased the sink capacity. Grain number per ear is vulnerable to environmental change and is closely related to male and female panicle differentiation and kernel abortion (Chen et al. 1988; Greenway et al. 2006). Chen et al. (1989) reported that 1000-grain weight decreased after being waterlogged at V3, grains per ear decreased after being waterlogged at V6, and 1000-grain weight and grains per ear decreased after waterlogging at VT. In addition, Yang and Chen (1998) demonstrated that waterlogging at the early stage of maize growth reduced grain rows per ear and grain number, while waterlogging at later stages decreased the 1000-grain weight. In the current study, waterlogged maize had prolonged growth stages and decreased ear length, grain rows per ear, and 1000-grain weight. Bald tip length and the proportion of bald increased after waterlogging. Waterlogging delayed summer maize growth processes and reduced the dry matter accumulation, harvested ear number, grains per ear, and grain weight to varying degrees. At V3, vegetative growth was dominant. At V6, both vegetative growth and reproductive growth of maize occur simultaneously. Summer maize photosynthetic performance and biomass, and the male and female panicle were delayed after waterlogging, leading to lower grain number and weight. At 10VT, summer maize grain filling began with the completion of vegetative growth and panicle differentiation. Waterlogging mainly decreased the dry matter accumulation and transportation from other organs to the grain, resulting in significant decreases in grain weight and yield.

Grain filling is crucial for grain yield in summer maize, and the grain filling rate is affected by both the genotype and environmental conditions. Dry matter accumulation in the filling stage was influenced by filling duration and rate (Li et al. 1999). Large storage capacity, rapid starch accumulation, and long filling durations resulted in high grain weight (Hu et al. 1995; Wu 2008). Our results suggest that waterlogging at V3 delayed growth processes and reduced dry matter accumulation, resulting in a lack of source material supplies when the grain sinks started to form. After waterlogging at V6, the vegetative and reproductive growth of summer maize was restricted and the sink capacity (as well as the source material supply) was limited. Waterlogging at 10VT mainly affected the grain formation and ear volume, leading to an insufficient sink capacity. Therefore, waterlogging seriously

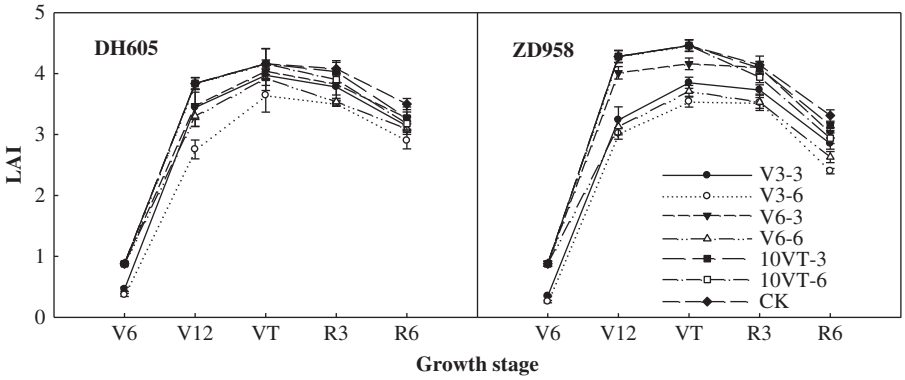


Fig. 2. Effects of waterlogging on leaf area index of summer maize.

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affected normal grain filling with Dmax, Gmax, Wmax, and P decreasing significantly in waterlogged plants, resulting in a decrease in grain weight.

Waterlogging first limited the growth of summer maize and decreased the photosynthetic area, the photosynthetic rate, and assimilation per unit area, thus affecting dry matter accumulation and the allocation ratio of various organs. Dry matter was mainly transferred to the vegetative organs during the early growth stages, and transferred to the ear and grain during the later growth stages to establish sinks (Shen et al. 2006). Hu et al. (2004) demonstrated that waterlogging in wheat significantly decreased dry matter accumulation in each organ and led to increasing proportions of stem and leaf dry matter distributions. Our results also demonstrate that dry matter accumulation at various growth stages decreased significantly compared with CK after waterlogging. The dry matter accumulation of V3-6 was the lowest among the treatments. Waterlogging at V3 and V6 markedly decreased dry matter accumulation before VT, but waterlogging at 10VT affected dry matter accumulation from VT to R6. The leaf and stem dry matter proportions increasedafter waterlogging. However, the proportion of the ear dry matter decreased. Waterlogging reduced dry matter translocations from leaves and stems to ears, resulting in a lack of sink capacity and inadequate grain filling.

A significant linear relationship existed between yield and plant morphology. Plant height affects the solar radiation distribution in the maize canopy. The taller plants may have greater canopy light distribution and the construction of high-yield groups (Pereira and Lee 1995). Plant height had a significant positive effect on corn yield (Chapman and Edmedes 1999), and LAI had an important role in economic production. Li et al. (2011) concluded that waterlogging did not significantly affect plant height. Chen et al. (1989) and Jiang et al. (2002) reported that the green leaf number and LAI were reduced after waterlogging. In this study, plant height of summer maize was significantly decreased after waterlogging at V3 and V6, which was inconsistent with the results of Li et al. (2011). The reduction of plant height after waterlogging was not conducive to solar radiation distribution in the maize canopy or the construction of high-yield groups. After waterlogging, the LAI significantly decreased. The reduction of LAI resulted in decreased photosynthetic area, and reduced accumulation and translocation of photosynthate, thus decreasing the grain yield.

CONCLUSION

Waterlogging under field conditions delayed the growth process of summer maize and reduced dry matter accumulation. The decrease in ear volume resulted in a lack of sink capacity. Because of the lack of sufficient source material supplies, the formation of grain sinks was limited, resulting in the disturbance of normal growth and development. Moreover, plants were most susceptible to damage when waterlogging occurred at V3, followed by V6 and 10VT; damage increased with increasing waterlogging durations.

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Araki, H., Hanada, A., Hossain, M. A. and Takahashi, T. 2012. Waterlogging at jointing and/or after anthesis in wheat induces early leaf senescence and impairs grain filling. Field Crops Res. **137**: 27–36.

Bange, M. P., Milroy, S. P. and Thongbai, P. 2004. Growth and yield of cotton in response to waterlogging. Field Crops Res. **88**: 129–142.

Belford, R. K. 1981. Response of winter wheat to prolonged waterlogging under outdoor conditions. J. Agric. Sci. **97**: 557–568.

Bhan, S. 1977. Effect of waterlogging on maize. Indian J. Agric. Res. **11**: 147–150.

Chapman, S. C. and Edmeades, G. O. 1999. Selection improves drought tolerance in tropical maize populations: II. Direct and correlated responses among secondary traits. Crop Sci. **39**: 1315–1324.

Chen, G. P., Zhao, S. X. and Liu, Z. W. 1989. Studies on waterlogging of corn and protection measures II. Responses of corn to waterlogging in various growing stages. Acta Agric. Boreali-Sin. **4**: 16–22.

Chen, G. P., Zhao, S. X., Yang, H. Y. and He, C. M. 1988. Studies on waterlogging of corn and protection measures I. Effects of waterlogging on bud bursting stage on the emergence and early growth of seedling of corn. Acta Agric. Boreali-Sin. **3**: 12–17.

Dickin, E. and Wright, D. 2008. The effects of winter waterlogging and summer drought on the growth and yield of winter wheat (*Triticum aestivum* L.). Eur. J. Agron. **28**: 234–244.

Greenway, H., Armstrong, W. and Colmer, T. D. 2006. Conditions leading to high CO_2 (>5 kPa) in waterlogged-flooded soils and possible effects on root growth and metabolism. Ann. Bot. (Lond.) **98**: 9–32.

Grzesiak, S., Hura, T., Grzesiak, M. T. and Pienkowski, S. 1999. The impact of limited soil moisture and waterlogging stress conditions on morphological and anatomical root traits in maize (*Zea mays* L.) hybrids of different drought tolerance. Acta Physiol Plant **3**: 305–315.

Howell, T. A. and Hiler, E. A. 1974. Effect of inundation period on seeding growth. Trans. ASAE **17**: 186–288.

Hu C. H. 1995. Corn cultivation physiology. Agriculture Press, Beijing, China. pp. 122–124.

Hu, J. C., Cao, W. X., Jiang, D. and Luo, W. H. 2004. Quantification of water stress factor for crop growth simulation I. Effects of drought and waterlogging stress on photosynthesis, transpiration and dry matter partitioning in winter wheat. Acta Agron. Sin. **3**: 315–320.

Jiang, D., Tao, Q. N. and Zhang, G. P. 2002. Effect of waterlogging on senescence of flag leaf and root of wheat Yangmai 5. J. Appl. Ecol. **13**: 1519–1521.

Krause, G. H. and Weis, E. 1984. Chlorophyll fluorescence as a tool in plant physiology. Photosynth Res. **5**: 139–157.

Li, J. C. 1997. Waterlogging physiology of wheat and its relation with wheat productivity. Plant Physiol Commun. **33**: 304–312.

Li, J. C., Dong, Q. and Yu, S. W. 2001. Effect of waterlogging at different growth stages on photosynthesis and yield of different wheat cultivars. Acta Agron. Sin. **27**: 434–442.

Li, S. C., Sheng, Q., Lu, J. H. and Meng, B. M. 1999. Growth analysis on the process of grain filling in maize. J. Shihezi Univ. (Natural Science) **3**: 1–5.

Li, X. Y., Liu, Z. Y. and Li, T. X. 2011. An impact test study of the flood disasters on summer corn's characters and yield. Acta Meteorol. Sin. **31**: 79–82.

Malik, A. I., Colmer, T. D., Lambers, H. and Schortemeyer, M. 2001. Changes in physiological and morphological traits of roots and shoots of wheat in response to different depths of waterlogging. Aust. J. Plant Physiol. **28**: 1121–1131.

Mckevlin, M. R., Hook, D. D. and Rozelle, A. A. 1998. Adaption of plants to flooding and soil waterlogging. Southern forested wetlands: ecology and management. Lewis Publishers, Boca Raton, FL. pp. 173–203.

Milroy, S. P., Bange, M. P. and Thongbai, P. 2009. Cotton leaf nutrient concentrations in response to waterlogging under field conditions. Field Crops Res. **113**: 246–255.

Musgrave, M. E. and Ding, N. 1998. Evaluating wheat cultivars for waterlogging tolerance. Crop Sci. **38**: 90–97.

Patwardhan, A. S., Nieber, J. L. and Moore, I. D. 1988. Oxygen carbon dioxide and water transfer in soils: mechanisms and crop response. Trans. ASAE **31**: 1383–1395.

Pereira, M. G. and Lee, M. 1995. Identification of genomic regions affecting plant height in sorghum and maize. Theor. Appl. Genet. **90**: 380–388.

Przywara, G. and Stcdaniewski, W. 1999. The influence of water logging at different temperatures on penetration depth and porosity of roots and on stomatal diffusive resistance of pea and maize seedlings. Acta Physiol Plant. **21**: 405–411.

Ritter, W. F. and Beer, C. E. 1969. Yield reduction by controlled waterlogging of corn. Trans. ASAE **12**: 46–50.

Shen, L. X., Wei, Y. P., Wang, P., Yi, Z. X., Zhang, H. F. and Lan, L. W. 2006. Effect of nitrogen supply on early kernel development and yield in summer maize (*Zea mays* L.). Acta Agron. Sin. **11**: 1746–1751.

Tan, W. N., Dai, T. B., Jing, Q., Cao, W. X. and Jiang, D. 2007. Effect of post-anthesis waterlogging on flag leaf photosynthetic characteristics and yield in wheat. J. Triticale Crop. **27**: 314–317.

Wu, C. S. 2008. Studies on characteristics of grain filling and dry matter accumulation of super high yield maize. J. Jilin Agric. Univ. **30**: 382–385.

Yang, J. P. and Chen, J. 1998. The effects of waterlogging at different growth stages on the growth and development of spring corn. Acta Agric. Zhe jiangensis. **10**: 188–193.

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